Desire Job Hayang

desire.hayang@aims-cameroon.org +237 695 900 562 LinkedIn GitHub
https://orcid.org/0000-0002-XXXX-XXXX

Professional Profile

Master's graduate with two master's degrees in Mathematics, with research backgrounds in algebraic geometry and combinatorics, and differential geometry. Strong foundation in mathematical reasoning, modelling, and problem-solving. Currently gaining hands-on experience in Data Science and AI through an internship at Quantum ARISE, contributing to ThermoFab, a scientific platform that applies data-driven and machine learning approaches to predict thermoelectric material properties. Developing practical skills in machine learning, scientific data analysis, computational modelling, databases, APIs, and software development. Combining mathematical expertise with computational experience to address complex scientific problems, with additional interests in DevOps engineering, design, and the development of intelligent and secure technological solutions.

Areas of Expertise

Professional Skills

- Teamwork and ability to work under pressure
- Scientific Problem Solving and Critical Thinking
- Scientific Writing & Academic Communication
- Data Science & AI
- Design thinking
- Software developer
- Agile Development
- Personal Development
- Drafting of entrepreneurial projects
- Mathematical Modeling
- Differential Geometry & Topology
- Real Analysis & Functional Analysis
- Linear Algebra & Abstract Algebra & Set Theory
- Numerical methods and PDEs

Technical Skills

- **Programming Languages:** Python, JavaScript, SQL, CSS, R
- **Data Science & Machine Learning:** NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch
- **Maths Software:** LaTeX, SageMath
- **Development & DevOps:** Git, GitHub, VS Code, Linux, Docker

Education

MSc in Mathematics *August 2025 – June 2026*
African Institute for Mathematical Sciences (AIMS), Cmeroon
Mastercard Foundation Scholar
Research thesis: Toric Fano varieties
Research area: Algebraic Geometry and Combinatorics

MSc in Mathematics *September 2026*
University of Ngaoundere, Cameroon

Research thesis: Prolongation of multiplicative Dirac structure related to Weil bundle
Research area: Differential Geometry

BSc Mathematics *August 2021*
University of Ngaoundere, Cameroon
Pure Mathematics

Professional Experience

Intern, Quantum ARISE
Lomé, Togo

- Contributing to the development of **ThermoFab**, a scientific platform for predicting thermoelectric material properties using data-driven and machine learning approaches.
- Working with **scientific data, machine learning models, computational methods, databases, and APIs** to support reliable prediction workflows and scientific analysis.
- Gaining hands-on experience in **Python, scientific computing, data processing, software development, and AI-driven scientific applications**.

Teaching/Academic Support Volunteer
University of Ngaoundere
November 2024 – March 2025

- Assisted undergraduate students with exercises in calculus, algebra, and topology.
- Created clear explanations to reinforce mathematical intuition and rigor.
- Helped students develop structured approaches to complex problems.

Tutoring and Review Sessions
Secondary High School, Gangassaou
November 2019 - June 2021 and *December 2023 - May 2024*

- Assisted learners with exercises in calculus, algebra, and arithmetic.
- Created clear explanations to reinforce mathematical intuition and rigor.
- Helped learners develop structured approaches to complex problems.
- Prepared students for official examinations.

Talk at AIMS-Cameroon
November 2025

- Presentation on the complexity of numbers π and e
- Demonstration of the irrationality and transcendence of the number e
- Demonstration of the transcendence of irrationality using Niven's approach (1947) for the number π
- Presentation on the importance of numbers pi and e in real life

End-of-Course Presentations
AIMS-Cameroon
November 2025 - April 2026

- Presentation on Ring, Group and Coding Theory and Cryptography
- Final course presentation on Applied Algebraic Geometry
- Presentation on Mathematical Problem Solving
- Final course presentation on Machine Learning and AI
- Presentation on R and Python programming languages
- Report on Actuarial Mathematics and Risks
- Debate on AI and Preservation of African Values
- Presentation on Numerical Methods for PDEs and Computational Fluid

BOOTCAMP AIMS-Cameroon
June - July 2026

- Design thinking session
- Power BI session
- Introduction to JavaScript

Projects

AlloDoc (2025)

- Creation of an online service for medical assistance
- Creation of a professional network with top doctors
- Participation in reducing disease rates through technological tools

Multi-Vendor E-Commerce Marketplace

- Implement Role-Based Access Control (RBAC)
- Streamline Product Management
- Unified Shopping Experience
- Deliver a Functional MVP

References

Available upon request.